What is the influence of knee joint movement on the maximal average mechanical power output of human quadriceps femoris muscle?

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Access the code and full analysis notebook at https://edwinreuvers.github.io/publications/hq-ampo

Abstract

Performance during motor activities such as wheelchair riding, cycling, rowing and speed skating, depends critically on the average mechanical power output (AMPO) produced by the muscles. To maximise short-duration performance, limb movements should allow muscles to deliver maximal AMPO. However, it is unclear which movement maximises AMPO of human muscle. In this study, we employed a Hill-type muscle-tendon-complex (MTC) model to predict the maximally attainable AMPO of human *m. quadriceps femoris* for various imposed periodic knee joint movements. Based on these predictions, we selected one set of conditions predicted to yield identical maximally attainable AMPO despite substantial variations in knee joint movements and another set of conditions predicted to yield substantial variations in maximally attainable AMPO. In the experiment, periodic knee joint movements were fully imposed by a knee dynamometer. Participants were instructed to maximise AMPO and, to this end, received visual feedback on their cumulative mechanical work throughout each cycle. Experimental data closely matched predictions derived from the Hill-type MTC model, confirming the validity of the model. Model predictions showed a substantial influence of knee joint movement on the maximally attainable AMPO. Specifically, predictions revealed a strong interaction between cycle frequency and knee joint excursion: increasing one necessitates a decrease in the other to maximise AMPO. Even more interestingly, *m. quadriceps femoris* should spend about 80% of the cycle duration while shortening, independent of cycle frequency and/or knee joint excursion.

1 Introduction

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Grieve et al. (1978)

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2 Appendix

2.1 Appendix 1

2.2 Appendix 2

3 Appendix

3.1 Appendix 1

References

Grieve, D. W., Pheasant, S. and Cavanagh, P. R. (1978). Prediction of gastrocnemius length from knee and ankle joint posture. In *Biomechanics VI-A* (ed. Asmussen, E.) and Jorgensen, K.), pp. 405–412. Baltimore: University Park Press.